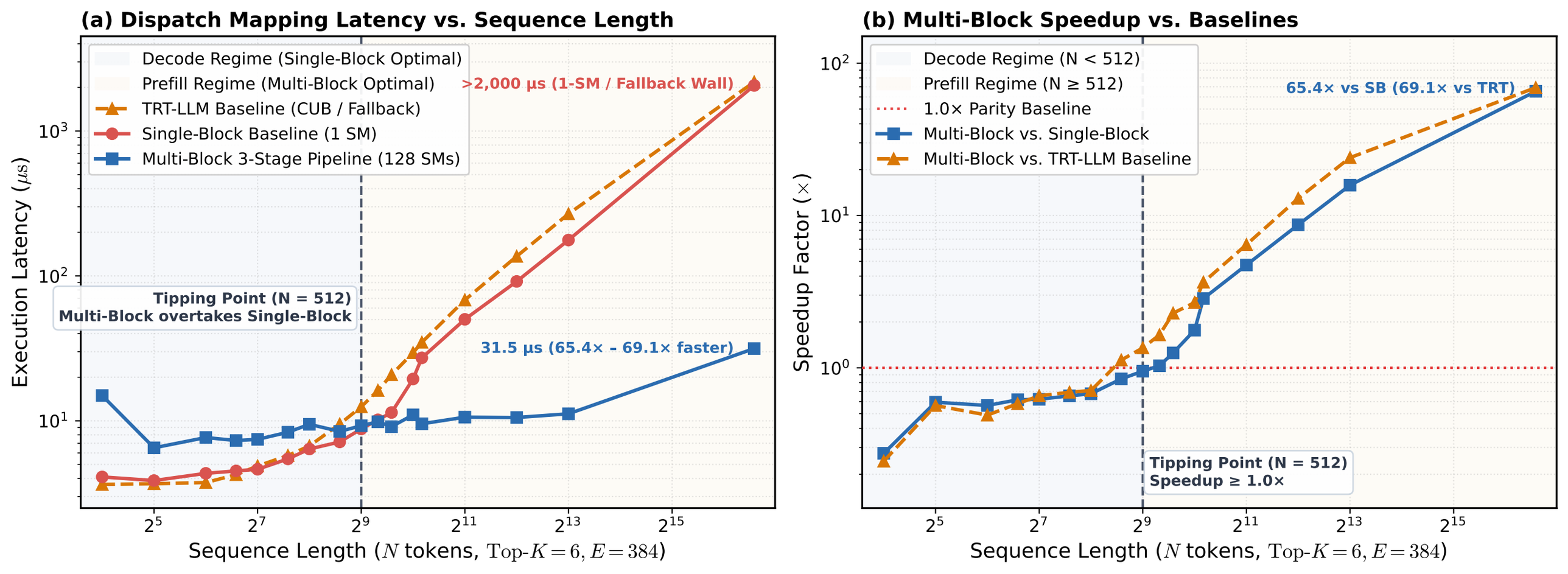
(a) Dispatch Mapping Latency vs. Sequence Length
Decode Regime (Single-Block Optimal)
Prefill Regime (Multi-Block Optimal)
TRT-LLM Baseline (CUB / Fallback)
Single-Block Baseline (1 SM)
Multi-Block 3-Stage Pipeline (128 SMs)
>2,000 µs (1-SM / Fallback Wall)
Tipping Point (N = 512)
Multi-Block overtakes Single-Block
31.5 µs (65.4× – 69.1× faster)
Execution Latency (µs)
Sequence Length (N tokens, Top-K = 6, E = 384)
(b) Multi-Block Speedup vs. Baselines
Decode Regime (N < 512)
Prefill Regime (N ≥ 512)
1.0× Parity Baseline
Multi-Block vs. Single-Block
Multi-Block vs. TRT-LLM Baseline
65.4× vs SB (69.1× vs TRT)
Tipping Point (N = 512)
Speedup ≥ 1.0×
Speedup Factor (×)
Sequence Length (N tokens, Top-K = 6, E = 384)